

The global access challenge

R P Searle, A R J Cook and D M Spirit

This paper looks at the access options and challenges facing the new operators in an increasingly liberalised telecommunications market. It considers the opportunities and risks associated with moving from a predominantly leased-line approach to one where a combination of owned and indirect access is also used. The business structures that can be used to create a new operator are considered, together with a high level view of the technology options they have. The fundamentals of implementing a network are also discussed.

1. Introduction

It is easy to forget that the origins of the vast and complex telecommunications industry we see today can be found in the state owned Post, Telegraph and Telephone (PTT) organisations that began in the early years of this century. Until relatively recently globalisation was about international connections between these national carriers. The customers, or subscribers as they used to be called, had little or no choice in either the products or services they could buy, or the price they paid.

New operators have existed for sometime, but generally they have only been permitted to provide 'non-core' services, such as radiopaging and packet switching. However, following events such as the forced break-up of the AT&T monopoly in the US and the privatisation of BT in the UK, other countries began to realise that there were other, perhaps faster, ways to facilitate the development of telecommunications.

In the early days of liberalisation, the competition tended to be in defined areas. For example, the long-distance carriers in the USA, namely AT&T, Sprint and MCI, were required to deliver their services to the end customers through local operators.

Over the past decade there has been a significant shift to full liberalisation. New companies can position themselves either as second operators or as niche businesses, targeting specific market segments and services.

Mobile cellular operators have been treated somewhat differently. From the outset there has been competition in most countries, including the UK, where two operators were licensed initially at 900 MHz, followed by a further two in the early 1990s. This approach has been partly due to the limited availability of radio spectrum. All these operators

are licensed to provide the network infrastructure and access the customers over a radio interface.

2. Why do second operators need to worry about access?

Having a properly planned, targeted and operated access network is key to the success of a second operator for a variety of reasons. Some of these reasons are connected with the operator's ability to differentiate itself from other operators, while others allow it to disengage its financial situation from some of the specific actions that could be taken by the incumbent PTT.

Second operators often entered the market by offering international data, domestic data and voice services to the corporate business community. They did this by connecting customers to their own switch sites using leased lines from the incumbent PTT or other in-country operators. Offering indirect access to the smaller businesses and residential communities has in some cases, followed this initial approach to large customers. Indirect access allows customers to make calls over the incumbent's access network, but breaks out to the second operator's network for long-distance transmission and switching. This is described in some detail by Mallett and Fletcher [1].

These two techniques for reaching customers are valuable tools in the second operators' armouries, enabling them, with a relatively small investment, to quickly reach both a wide customer base with indirect access, and a valuable customer base with the corporate services. Among other benefits, this activity also enables the new company to establish its brand identity within the country.

However, there are also significant downsides to using this strategy as a long-term solution:

- the new operator's quality of service and responsiveness is locked to the best that the incumbent can provide to any operator,
- the new operator's services are limited by what the incumbent can provide,
- the new operator's business is sensitive to the pricing strategies of the incumbent — any operator running such a network has to hand over a large portion of its revenues to the incumbent operator, either in access interconnect charges, or in rental for leased lines.

In order to free it from these constraints and also to retain a larger proportion of the revenues, the new operator must reduce the amount it pays to the incumbent operator by, for example, bypassing the incumbent's access network.

However, an access network, whether leased or wholly owned, is probably the largest long-term financial commitment that any new entrant will need to make. Because of this, it is absolutely essential that an owned access network is both properly planned and targeted, since mistakes in this area are likely to have a 'make or break' impact on the profitability of the new company. It is generally not appropriate for a fledgling operator to rush into building a homogenous network across the whole of the country to serve all customer segments. Rather, the new operator should consider all the options open to it and use them to build a patchwork of different solutions to serve appropriate parts of the market.

Once an entrant has made the decision to build their own access network, there are a number of new opportunities open to it including:

- the ability to offer a quality of service which is not dependent on the incumbent operator's response times or quality criteria,
- the ability to launch new services more quickly — in particular, this positions the new operator much better in its ability to respond to the 'datawave' or to provide other new broadband services,
- the flexibility to offer services at prices which are not dictated by the incumbent's pricing policy, but by the new operator's own cost base — an example of this could be that a second 2 Mbit/s connection could be offered at a substantially lower price than the first connection since the first connection could be priced to bear the majority of the up-front network build costs,
- the customer is more likely to feel 'owned' by the new operator, rather than shared across two or more operators,

- operators with both fixed and mobile networks have the opportunity to develop services exploiting fixed/mobile convergence (FMC) [2], such as integrated billing, automatic routing of calls to either mobile or fixed terminals, etc, in order to provide added customer value.

3. The access jigsaw

There are a wide variety of options open to any operator for owned access, but there is not a simple formula that will provide a single answer for use in all circumstances. The choice an individual operator makes will be based on a number of factors including:

- the anticipated market in that country,
- the price structures offered by the incumbent for both wholesale and retail services,
- the advantages of using any facilities owned by partners in the company (e.g. a power utility may provide access to an existing physical network which can be reused),
- availability and pricing of leased copper and/or fibre from other sources,
- availability and pricing of access bandwidth leased from other operators,
- the scope of the operator's complete network build activity, e.g. could a general-purpose access network utilise the backhaul network for mobile base stations?

What is essential is that, because of the high cost of building any owned access network, whatever options are chosen, the geography and market in the area where the access network is being built must be well understood. There should also be a high degree of confidence that there will be sufficient customers to achieve an adequate level of fill on the deployed assets. As with all planning activities, there must be an effective strategy in place to deal with areas where there is more demand than was originally anticipated.

With this in mind, the individual technology options open to a new operator are each applicable across a limited number of market segments. There are many ways of cutting the market, but one approach is to break it into four main segments:

- corporate/multinational corporation (corporate/MNC) — large customers, often with many sites, some of which will have 100 + lines,
- small/medium enterprises (SMEs) — small and medium sized business customers, down to perhaps 5 lines,

- small office/home office (SOHO) — very small business customers and home-based businesses,
- mass market — low usage business customers and the top-end residential market.

Within these segments, the broad applicability of the various technology options can be shown in Fig 1, which shows the scale of the problem that a new entrant faces in developing a workable access strategy with this plethora of technical options available to them and with all manufacturers keen to push their own solutions forward. However, over many years, BT has developed a wealth of knowledge on these technologies. More detail on many of the technology options is available elsewhere in this edition of the Journal, but this paper describes the broad applicability of them to new operators.

3.1 Fibre options

Both in the UK and overseas, fibre is the technology of choice for delivering fixed services to customers because it gives the maximum flexibility, has almost unlimited upgrade capability to meet future needs and it can provide top quality service to the customer. There are two broad types of fibre transmission systems — point-to-point and PON (passive optical network) based systems.

Point-to-point systems can deliver very high bandwidths to the customer using a dedicated fibre or pair of fibres between the new operator's point of presence and the customer. The use of PON [3] can improve the economics of direct fibre to customers by sharing some parts of the fibre network itself and the exchange-end electronics across a number of customers by use of a passive splitter. Currently, this solution is applicable to those customers requiring from around 1 Mbit/s up to a few tens of Mbit/s, with the upper limit being imposed by the fact that a PON shares its capacity across a number of customers. Within the Full Service Access Network (FSAN) body, there is work to standardise the emerging generation of ATM PON equipment for broadband services [4]. This world standard equipment will be equally appropriate for use overseas as it will be for use within the UK.

In all cases, however, fibre can be costly to install, since it requires extensive civil works to be carried out. Deployment will therefore be limited to districts where there is a very high probability of getting a good take-up of high value customers. Into these areas it would be normal to follow the deployment practice shown in Fig 2.

This approach builds a fibre spine or ring into an area which will have a high density of corporate/MNC/SME customers, e.g. business centres in a city. Short spurs are then installed off this spine/ring to provide the 'all

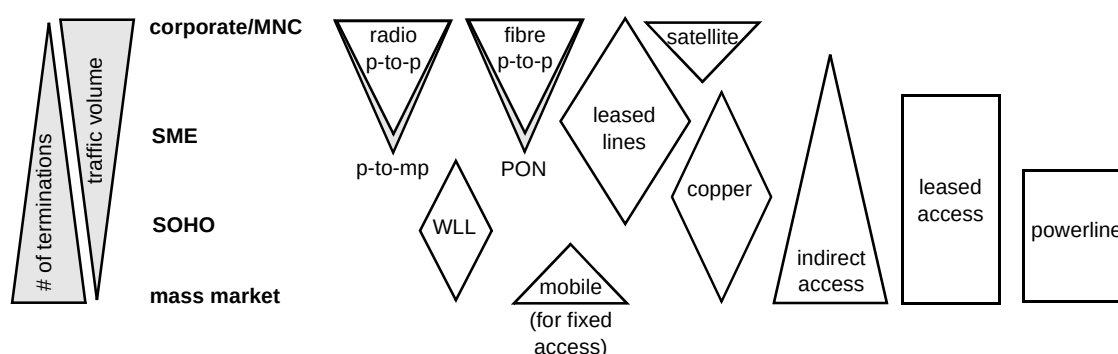


Fig 1 The various technology options.

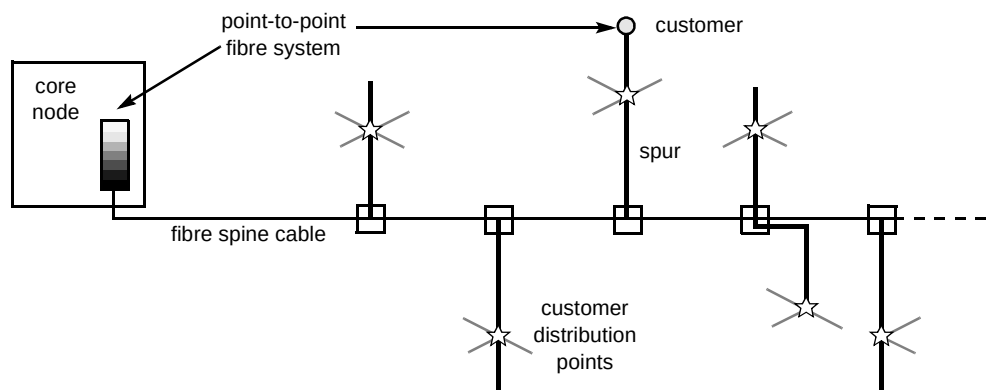


Fig 2 Deploying fibre into the access network.

important' feeds to individual customers. This technique shares the bulk of the 'digging' cost across all the customers, and minimises the per-customer expenditure. A factor, which may make direct fibre more economic in some locations, is the availability of dark fibre from other operators, or from city utilities.

3.2 Radio

In the UK, BT's widespread use of radio systems in the access network is somewhat limited by regulation, and by the fact that it has an extensive wired network in place. However, other operators are using or are considering the use of radio to a much greater degree and there are two main options open to them — point-to-point, and point-to-multipoint radio.

Point-to-point radio

Point-to-point radio is the traditional and well established option, and, outside the areas where it is economic to provide direct fibre feeds, it offers a viable approach to owned access for the largest customers. Operators with a mobile radio network as part of their business will have the potential to share radio sites and 'backhaul' bandwidth, thereby providing a large number of radio PoPs (points of presence) close to the customers and making it easier to get a high coverage of line-of-sight paths.

Like point-to-point fibre systems, radio systems are capable of delivering a wide variety of services, and when planned properly can provide a very high grade of service.

Multipoint radio

Multipoint radio systems, particularly those offering broadband capabilities, are relative newcomers to the market and are still in development to one degree or another. They fall into two broad categories.

- **Wireless local loop (WLL)**

Used for delivery of up to narrowband ISDN rates, a good example, in the UK, is the system used by a new UK operator at 3.5 GHz delivering 32 kbit/s access that has the potential to be aggregated for higher rates. This category also includes DECT (digitally enhanced cordless telecommunications) based systems.

- **Wideband and broadband point-to-multipoint radio**

These are higher bandwidth systems providing a few hundred kbit/s up to a few tens of Mbit/s per customer depending on the system.

For a new operator, the higher bandwidth systems are generally more appropriate, being better suited to business customers' needs and having a better evolution path for these customers. The main advantages of multipoint radio systems over point-to-point ones are:

- reduced cost per customer, provided that sufficient fill can be achieved on the base station — again a good knowledge of the customer base in the target area is essential,
- multipoint systems require less tower space at base station sites — this may be an issue where a large number of customers are being served by radio from a single PoP.

More detail on radio systems is provided in Merrett et al [5].

3.3 Copper

In the UK, new operators, unlike BT, will not generally have access to copper of their own, and nor will they be likely to build much new copper. However, following deregulation, particularly in Europe, a number of countries are looking at offering 'unbundled access' to the incumbent's copper pairs at the MDF (main distribution frame). This gives the new operator some unique opportunities to lease copper infrastructure and use it to provide both basic and higher rate services using xDSL technologies [6]. However, it is in everyone's interest that robust agreements are put in place between the owner of the copper and all users of the network, in order to ensure that quality of service targets can be met. This is both from an operational viewpoint, and from one of ensuring interference-free co-existence between different operators' equipment inside the same cable bundles.

3.4 Indirect access and leased lines

As described earlier, these technologies, while not being owned access, will continue to be important to new operators for reaching particular customers. These may include:

- where a customer is not close to the fibre spine and has no line-of-sight radio path to a PoP,
- where a customer is in an area not covered by the new operator, e.g. local branches of a large corporate customer,
- where high customer churn is anticipated,
- where customer bandwidth is too low to justify the provision of owned access to them,
- mass-market entry using 'smartbox' technology.

3.5 Other solutions

There are other solutions, such as satellite, the use of mobile to serve fixed customers, powerline transmission systems, leasing access from other operators, etc, which will also be appropriate under particular circumstances. However, unless and until the economics of some of these options change, they are likely to provide niche solutions rather than the major options described above.

4. Strategies of the new operators

One of the main objectives of liberalisation was to improve customer service and lower prices by creating competition to the incumbent operators. A natural reaction of PTTs was to look outside their own countries for opportunities to establish new businesses to compete with the 'local' incumbent operators.

Global voice and data services for corporate customers were the markets that several PTTs and large US operators decided to target in the early 1990s. BT and MCI formed Concert and France Telecom, Deutsche Telecom and Sprint, the US long distance carrier, formed Global 1. Their networks were built by connecting their switches with national and international leased bandwidth. The access parts of the networks were also formed from leased lines, rather than 'built' access.

Having moved into the provision of global services, BT and similar companies began to look into the opportunities offered by liberalisation in the domestic markets outside their own countries. With any industry, there are various ways to constitute a company and enter a particular market. It is probably true to say that in telecommunications most approaches have been tried somewhere in the world.

Many of the new operators have built their businesses on switching and a fibre core network, but still relied on the incumbent PTTs to deliver their services to the end customers. The cost and logistics of building a core network meant that established 'utility' companies from the gas, electricity or water supply industries found that it was worthwhile to diversify into telecommunications. Over the past few years cable TV companies have also extended their coverage significantly and, with increasing liberalisation, are able to offer telecommunications, as well as entertainment services. Some natural partnerships were made between the cable companies and the new 'utility-based' operators to provide end-to-end service using a high percentage of owned infrastructure.

The various approaches to forming these new companies tend to fit into a few categories.

5. Totally owned subsidiaries

In theory, a totally owned subsidiary could be established in all the countries of interest. This approach has the advantage of control, in terms of business and market strategies, of the new operator and is not subject to the aspirations of another company. Where the new operator is only delivering global services, e.g. for Concert, subsidiary companies operating as in-country agents have proved to be an effective approach. However, for domestic operations and the building of infrastructure, there is a much greater need for 'local knowledge', particularly for dealing with the regulator, negotiating with and managing suppliers, etc. This approach can also be very time consuming.

6. Joint Ventures

The Joint Venture (JV) offers a potential 'fast-track', in that by choosing the right partners, much of the local knowledge is obtained very quickly. Because a medium size company, with substantial backing, in terms of resources and expertise, can be established, it becomes viable to consider a wide range of products in competition with the local PTT and other operators.

These JVs will often have 'second operator' licences (both fixed and mobile) and will offer both international and domestic services. In many cases there will be an infrastructure 'partner' who has access to railways, cable TV networks, etc, together with a partner from, for example, a financial or retail outlet. In some cases another telecommunications operator will be part of the venture. An example is Viag Interkom in Germany, which is a joint venture between BT, Viag, the German industrial group, and Telenor, the Norwegian PTT. Viag has both industrial strength and reputation, as well as access to a significant amount of fibre-based core network in Bavaria. BT and Telenor, being major operators in their own countries, can of course bring substantial expertise to the business.

Start-up JVs will have different asset bases, depending on their partners. However, they do have many things in common, not least of which is the need to reach their customers in the most effective way. The investment decisions that the JVs make on the access aspects of reaching their customers will be based on the business and marketing strategies, combined with the current and evolving technologies. Many PTTs will have a wealth of experience in access technologies which gives both an understanding of the incumbent operators whose markets are now being threatened and the ability to support the JVs in developing their businesses.

7. Acquisitions

A joint venture, by definition, has various partners who influence the new company through their shareholdings. There are cases, particularly for the mobile business, where acquiring an existing cellular operator may offer the quickest way into a particular market sector. This approach does have risks, in that the acquired company will need to be integrated into the new organisation to some degree. This could include the operational aspects, product sets or technical strategy. However, for an established JV that wants to expand its product range and increase its customer base rapidly, this approach can be very attractive.

8. Shareholder

A variation of acquisition and joint venture is to become a shareholder in an existing company. This differs from a JV in that the operator is normally operational rather than in start-up. An example is Cegetel in France in which

BT has a shareholding within a complex set of partnerships and other shareholdings which include SNCF, the French railways, Vodafone and the Southwestern Bell Company from the USA (SBC). The company, particularly the cellular business of SFR, was already established when BT became a shareholder.

9. Regulatory issues

In the past, the regulations protected the incumbent PTT by excluding other operators from offering anything but 'peripheral' services. Somewhat ironically, the regulations now tend to favour the new operators, by ensuring that the incumbents do not abuse their position as the dominant carriers.

In Europe, the 1 January 1998 saw a major step in the liberalisation of telecommunications across the European Union. Although there are differing approaches between countries, the end result will be one of the most open markets in the world.

10. Unbundling

'Unbundling' is a generic term used to describe the partitioning of each element of a telecommunications network (normally the incumbent's), such that each component part could be leased, or even sold to another operator. Taken to extremes this could result in parts of switches and the associated processors being leased to a third party. Figure 3 shows just some of the items that could be unbundled. In terms of access, unbundling of the local loop is of direct interest, particularly with the increasing deployment of DSL (digital subscriber line) systems [6].

Where the regulations allow unbundled copper¹ new operators will, potentially, be able to use DSL systems to provide a range of new services, for example high-speed Internet access. The unbundling of copper could have a major impact on the way the new operators provide access to their customers.

¹ The term 'dark copper' can be interchanged with the terms unbundled copper, 'dry copper' and 'MDF Interconnect'.

The technical challenges with dark copper¹ will broadly be the same in all countries. However, there may be differences in the type of cables that were installed originally, the topology of the network and condition of the cables. It is also possible that aluminium could exist, dating from the days when copper prices reached an all time high and alternative solutions were used.

One of the most important areas to address for a new operator is the service level agreement (SLA) for dark copper with the incumbent operator. This is put in place to clarify the relationship between the incumbent and the new operator and to reduce the risk of either party damaging the performance and quality levels to their respective customers. Both parties also need to be clear about agreements on such things as accommodation, cable records, the time to repair faults and access to test systems.

11. Interconnect

Even with global agreements and networks, new operators are extremely unlikely to reach the point where they are providing direct access and end-to-end routing to all their customers. This implies that they will need to interconnect with the incumbent PTT or other operators. Some of the interconnection will be within the core network where traffic is handed over between networks at points of presence.

From an access perspective interconnect can occur at several points. For example, with indirect access (also known as trunk bypass) the operator interconnects on the core network side of the incumbent's switches and routes long distance traffic over their own core network. The interconnect agreement would cover various areas, including charges, billing and operational issues, such as mean time to repair and alternative routings. There will also be a range of technical issues, e.g. signalling, delay and echo performance, that will need agreed parameters and a testing regime.

Dark copper¹ is another area that requires a well-defined interconnect agreement as there are additional issues, such as accommodation access and power supply integrity, that

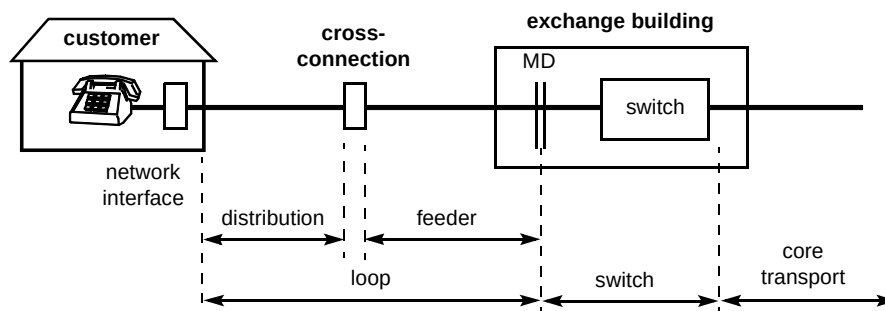


Fig 3 Some of the components that could be 'unbundled'.

require non-ambiguous agreement to ensure that the overall service levels are maintained.

12. New operators' approaches to access

Outside the USA, many of the new operators have established themselves in Europe (see Fig 4). One of their biggest challenges is to be able to reach their customers efficiently, by a combination of technologies, either leased or owned.

As you might expect there are similarities in approach between the ventures, but the local conditions do affect the particular implementations. Some examples are given below.

- Telfort

Telfort, which was launched in the Netherlands in 1996, now has a rolling programme of installing fibre networks to the 'Business Parks' across the country, where many of the large corporate customers and some of the small-to-medium size companies are clustered. BT's partner in Telfort is Nederlands Spoorwegen, the Dutch railway company, who already own several thousand kilometres of fibre installed alongside the railway tracks close to many of the business parks.

Point-to-point radio is also an option for providing service rapidly to new customers and as a longer term alternative for customers who are not close to the installed fibre rings.

- Cegetel

In France, Cegetel, which is owned by Compagnie Generale des Eaux, is, as explained above, a complex set of shareholdings and relationships. SNCF, the French railway company, already have a fibre network throughout much of France. Unlike most ventures, Cegetel already has an established mobile operator, SFR, with over 2.5 million customers via a GSM network covering 90% of the French population and a paging business with nearly 500 000 customers. This means that they are already firmly established in the mass market.

In terms of fixed access Cegetel is investing in its own fibre infrastructure particularly in the main conurbations such as La Defense in Paris, where there are already a significant number of access points to the Cegetel network. With the fibre in place, decisions are being made on how best exploit the capacity. Point-to-point fibre systems offer the most capacity, but PONs would enable more customers to be served from the same fibre pair.

In addition SFR already have a network of microwave links to feed their cellular base stations. The access strategy will therefore take account of these existing assets as well as the need to reach the growing customer base in different market sectors.

- Clear Communications

Clear Communications is New Zealand's second largest operator. Clear has always positioned itself as a company with service excellence. One of the ways it achieved this is by managing services and the network

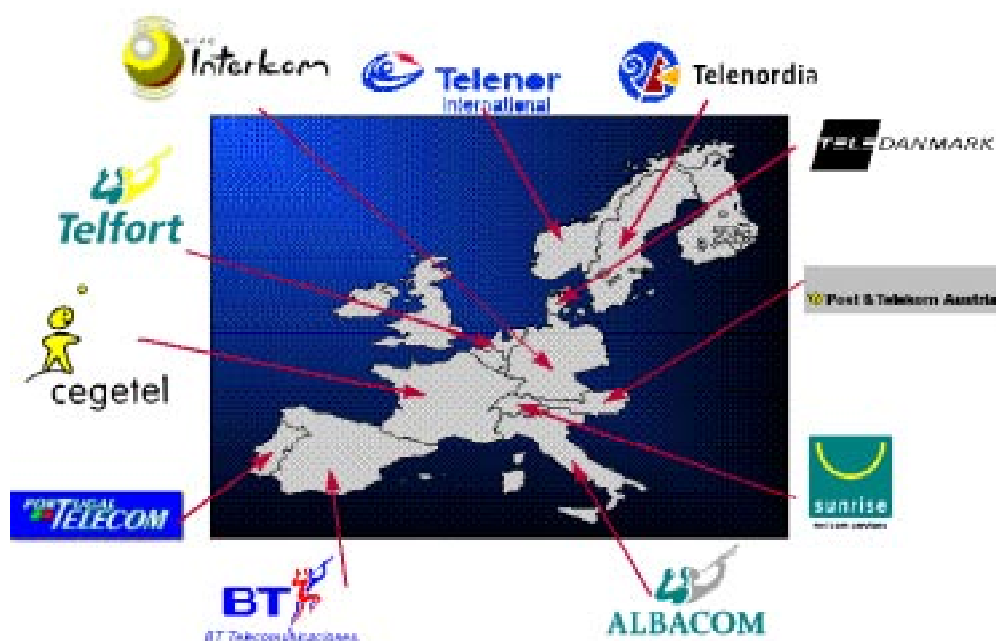


Fig 4 Some of the new operators and BT's partners in Europe.

right to the customer. Combining this objective with the delivery of new services, particularly broadband, has lead Clear to deploy their own fibre access into customer premises.

As already mentioned, fibre provides a future-proof way to access customers in that it has the potential to carry broadband services for the foreseeable future. However, it is expensive to install and therefore requires a targeted build, which relies on a good knowledge of both the customer locations and their likely service take up.

Many of the other technologies summarised in this paper and detailed elsewhere in this journal, e.g. point-to-point and point-to-multipoint radio, are being considered by the new operators.

13. Implementing the network

In some respects, the constraints within which a new entrant must operate when implementing a network are identical to those experienced by the incumbent operators. However, there are also major differences, the most significant of which is the scale of the access network deployment relative to the size of the new entrant's organisation. As already noted new operators will not, in general, be inheriting owned access networks. They will therefore need to build their access networks from scratch. Similarly, they will not have an operations and maintenance organisation of significant size. This leads to the need to consider outsourcing significant parts of building and running the network that an incumbent would traditionally do themselves. These functions may include areas such as the development of planning rules, negotiation with authorities for permissions to lay new duct, civil works, external plant, plant maintenance, infrastructure maintenance, access network management and overall project management. While these skills can be built within new network operators, they will inevitably take time to develop which would slow the rate of deployment of an owned access network.

The scope of a new entrant's outsourcing activities will vary on a case-by-case basis. In some cases, it may be that the new operator has plans to develop in-house only those areas which it feels are absolutely essential (e.g. a high level access strategy, leading into the development of planning rules). Industrial strength equipment vendors possess a range of capabilities in the majority of the areas mentioned above within their companies, in addition to being able to provide a wide range of technologies from which access networks may be constructed. While some of these functions could be provided centrally within the telecommunications equipment companies, for other areas, it is essential that a substantial team of people exists within the country.

In an extreme case, it is possible that a new operator could outsource the complete access network plan, provide, build, maintain and operate functions to an integrator, who is most likely to be one of the major equipment manufacturers. One of the key criteria in the selection of such a supplier is their ability to mobilise a suitable team of people within a country to provide this 'turn-key' solution. Innovative financing may also relate payments to revenue or profit of the operator, which would partially remove the negative effects of high start-up costs for an owned access network.

14. BT — working with its JVs

The JVs operate as independent companies from mainstream BT in order to allow them to compete effectively in their own markets and to give them a small business focus. However, they do this with the back-up of BT's resources behind them. This close focus on their own objectives, combined with the geographical spread of the JVs, means that achieving the close working between the JVs themselves and BT, requires positive management steps to be taken.

Having a number of 'friendly' new operators in slightly different phases of their life cycles provides a great opportunity to share the inevitable lessons learnt across the alliance.

Through the use of intranet technology, a shared workspace has been provided to which BT and all its Joint Ventures have access. As an adjunct to regular meetings of the JVs' and BT Global's senior management, this site provides an area in which the JV people at a working level can discuss and debate common problems and experiences, learn from one another and share technical information. It is also used to provide a 'textbook' of technical information on all aspects of access including technologies, operational issues, costs, performance benchmarking and the impact of access on the JV business as a whole.

15. Conclusions

This paper has looked at the growth of new operators in an increasingly liberalised market. The technologies available are of course the same as those used by the incumbent PTTs. However, the new operators will not necessarily address the same markets as the incumbent as they evolve rapidly from a relatively small base. This will result in them developing and implementing strategies that undermine the competition by using the available technologies in innovative ways. What is certain is that these new operators will play a major part in the continued and rapid growth of global telecommunications.

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Since joining BT in 1966, Roger Searle has worked on many aspects of transmission, particularly microwave radio, ranging from maintenance, through planning to the development of millimetric radio links. As Senior Radio Adviser for BT he led the BT delegation to the World Administrative Radio Conference (WARC) 92 and was closely involved in the development of spectrum requirements for UMTS. His career broadened in 1992 when he was Programme Manager for the development of traffic and signalling management systems in BT. Since 1996 he has been senior adviser for the JV

Alliances. This has involved working in Viag Interkom on the FMC (Fixed/Mobile Convergence) design and managing the Global Access project that has developed a major part of the WWW site used by the BT Alliance partners.



Andy Cook joined BT in 1988 and spent a number of years working on the development of the first generation of fibre based broadband network demonstrators. He then spent some time developing possible strategies for the use of TPON in the BT network, before moving back to broadband in order to build solution designs.

For the last two years, he has led a team of people providing business case and transport design support into the UK broadband programme and into the JVs through both generic and JV-specific activities. He also spent some months working on the access part of the FMC design for Viag Interkom, and has most recently been responsible for the technical content of the TABS WWW site.



Dave Spirit received a PhD in Physics in 1984 from Imperial College, London. Following two years as a researcher at Imperial College, he joined BT Laboratories in 1986 to work on the early development of optical fibre amplifiers.

He spent a number of years in the area of optical transmission and networking, which included the first collaborative demonstration of 20 Gbit/s transmission in the BT network in 1994.

In 1995, he joined a team built up to provide network design expertise for BT's Joint Venture companies. This has encompassed voice, data, access and core transport networks, particularly a pan-European network to be constructed by connecting JV networks together.

He is a Member of the Institute of Physics and a Chartered Physicist, and is co-editor of 'High Capacity Optical Transmission Explained'.